Coverage-Qualified Evaluation-Contract Auditing for Aerial Vehicle Detection

Xinling Wen, Jiahao Zhang, Wenqi Liu, Yangxu Ren, and Yu Chen

***Abstract*—Aerial-detection scores can move because a support rule changes both the target and the treatment of detections near excluded objects. We introduce a coverage-qualified evaluation-contract audit comprising a coverage gate, support measurement, an ordered contract path, and common-coverage rank inference. The path $A \to F \to R \to S$ changes one evaluator factor at a time and attaches an exact TP/FP/FN ledger to target filtering, removed-object neutralization, and source-ignore handling. At 24 pixels, VisDrone, UAVDT, and AI-TOD retain 73.3%, 95.1%, and 9.7% of valid vehicle boxes. Their fixed-baseline $A \to F$ micro-F_1 changes are $-.036$, $-.009$, and $-.330$; neutralizing removed objects then recovers $.067$, $.004$, and $.080$. AI-TOD covers 66.7% of images but only 28.2% of boxes, so its metric replay is conditional. On 548 common VisDrone images, 10,000 paired resamples of 76 sequences favor Y11m over ASD for fixed- and optimized-confidence F_1 at both IoU .25 and .50. A localization-by-metric control finds that AP25 favors Y11m at 24 pixels but is unresolved at .015, whereas AP50 is unresolved at 24 pixels and favors ASD at .015. A sequence-resolved ASD-favoring normalized conclusion appears only when stricter localization and score integration are combined.**

***Index Terms*—Aerial detection, annotation support, evaluation contract, coverage qualification, cluster bootstrap.**

I. Introduction

A detection score is interpretable only after its evaluation target and matching contract are fixed. This requirement is acute in aerial imagery: a common pixel cutoff can retain almost every vehicle in one source and remove most vehicles in another. Predictions near removed objects may then be counted as false positives or treated as neutral, while source-provided ignore regions add a separate decision. These choices enter the metric before detector outputs are compared.

Cross-dataset bias, heterogeneous label spaces, annotation reliability, and detector error taxonomies are well studied [1], [2], [3], [4]. Tiny-object procedures can also alter the prediction process itself [5]. This study instead measures evaluator effects after annotations and prediction caches are fixed, attributes those effects to explicit contract steps, and separates them from sampling uncertainty. Recent GRSL work likewise shows that validation conditions can materially affect remote-sensing conclusions [6], [7], but aerial-detection reports rarely connect coverage, support, contract semantics, and rank inference in one record.

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Three weaknesses make an unstructured sensitivity sweep hard to defend. First, incomplete prediction coverage can silently change the evaluated population. Second, endpoint ranges do not identify whether movement follows target filtering, the treatment of removed objects, or native ignore regions. Third, a fixed-confidence F_1 comparison can confound detector ranking with score calibration, while image-wise resampling can ignore sequence dependence. We address these weaknesses without training another detector.

Our contributions are fourfold. 1) We define a coverage-qualification gate that separates full-dataset annotation support from prediction-covered metric inference. 2) We introduce an ordered evaluation-contract path with an exact count-transition ledger. 3) We replay the same path on VisDrone, UAVDT, and a coverage-conditioned AI-TOD artifact [8], [9], [10]. 4) We qualify a frozen VisDrone pool with sequence-paired intervals and a localization-by-metric control that separates valid-match IoU from threshold optimization and score integration.

II. Coverage-Qualified Contract Audit

A. Gate 0 and support coordinates

We use VisDrone2019-DET validation, the UAVDT test split represented in COCO format for a common evaluator, and AI-TOD validation. The available AI-TOD reconstruction includes xView-derived imagery [11]. Vehicle mappings are VisDrone {car, van, truck, bus}, UAVDT {car, truck, bus}, and AI-TOD {vehicle}. Raw coverage is established from prediction image names *before* class filtering; a covered image with no retained vehicle prediction is an empty prediction, not a missing observation.

Gate 0 records the annotation denominator, raw coverage, and covered–noncovered differences before any metric is interpreted. Complete artifacts can enter all later layers. An incomplete artifact may enter support diagnosis and a clearly conditioned metric replay, but not a full-benchmark rank claim. We compare AI-TOD coverage groups by boxes per image, support, aspect ratio, and normalized nearest-neighbor spacing; standardized mean differences and Wasserstein distances are released.

Formally, let $\mathcal{I}_s$ be the declared annotation-image set for source s and $\mathcal{P}_k$ the image identifiers present in candidate artifact k. Its raw coverage is

$$q_{sk} = \frac{|\mathcal{I}_s \cap \mathcal{P}_k|}{|\mathcal{I}_s|}. \tag{1}$$

Annotation support is estimated on all of $\mathcal{I}_s$ and is independent of predictions. A metric endpoint for k is a full-source estimand only when $q_{sk} = 1$; otherwise it is written conditionally on $\mathcal{J}_{sk} = \mathcal{I}_s \cap \mathcal{P}_k$. A pairwise rank estimand uses the explicit common set $\mathcal{J}_{sk\ell} = \mathcal{I}_s \cap \mathcal{P}_k \cap \mathcal{P}_\ell$. This separation prevents a high-coverage artifact from silently redefining the benchmark denominator and prevents unmatched image sets from entering a detector comparison.

For a box (w, h) in image (W, H), absolute and normalized max-side support are

$$s_a = \max(w, h), \qquad s_n = \max\left(\frac{w}{W}, \frac{h}{H}\right). \tag{2}$$

We use the prespecified support grids $s_a \in \{16, 20, 24, 28, 32, 40, 48, 64\}$ pixels and $s_n \in \{.005, .0075, .010, .015, .020, .030\}$; 24 pixels and .015 are the headline interior values. Under uniform resizing by $\alpha > 0$, $s'_a = \alpha s_a$ whereas $s'_n = s_n$. Thus normalized support is resize invariant and absolute support is scale covariant. This property does not imply semantic equivalence across sources; ontology, acquisition geometry, cropping, and density can still differ.

Proposition 1 (scale behavior). Under the uniform transformation $(w, h, W, H) \mapsto (\alpha w, \alpha h, \alpha W, \alpha H)$ with $\alpha > 0$, normalized max-side support is invariant and absolute max-side support is covariant. This follows directly because $\max(\alpha w, \alpha h) = \alpha s_a$, whereas both factors of α cancel in s_n. The proposition motivates reporting both coordinates; it does not make a shared normalized threshold a claim of semantic equivalence.

B. Ordered contract-path decomposition

Let G be mapped valid boxes, $G_t \subseteq G$ the boxes retained by support rule t, $D_t = G \setminus G_t$ the removed boxes, and H the source-provided ignore regions. We declare four contracts as ordered pairs of scored valid and neutral regions:

$$\begin{aligned} \mathcal{C}_A &= (G, \varnothing), & \mathcal{C}_F &= (G_t, \varnothing), \\ \mathcal{C}_R &= (G_t, D_t), & \mathcal{C}_S &= (G_t, D_t \cup H). \end{aligned} \tag{3}$$

A scores all valid boxes with ignore protection off. F filters the target and leaves detections on removed objects eligible as false positives. R keeps the same filtered target but makes removed objects neutral. S then adds source-provided ignores. In every contract, detections are processed by descending score, retained valid GT is matched first, and only an unmatched detection can be neutralized. Ignore overlap follows the COCO crowd denominator, intersection over prediction area, at .5 [12].

For metric m at rule t, define adjacent path changes

$$\begin{aligned} &\delta_{\text{target}} = m_F - m_A, \qquad \delta_{\text{removed}} = m_R - m_F, \\ &\delta_{\text{source}} = m_S - m_R. \end{aligned} \tag{4}$$

They satisfy the exact path identity

$$m_S - m_A = \delta_{\text{target}} + \delta_{\text{removed}} + \delta_{\text{source}}. \tag{5}$$

The associated ledger records the exact ΔTP, ΔFP, and ΔFN at each step. Because F_1 is nonlinear, these are path-specific evaluator transitions, not order-free causal effects. The deterministic contract range $C_m(t) = \max_{c \in \{A,F,R,S\}} m_c(t) - \min_c m_c(t)$ remains a compact summary, but it is not a confidence interval.

The path also imposes implementation invariants. Each detection can match at most one scored valid box; scored-valid matching precedes neutralization; and a detection neutralized by D_t or H is neither TP nor FP. For micro-$F_1 = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN})$, the ledger is regenerated from image-level integer counts and must close both the count totals and the endpoint identity above. These checks distinguish a semantic contract change from accidental rematching or denominator drift.

C. Metric- and cluster-qualified rank protocol

The metric layer replays one frozen baseline prediction artifact per source at confidence and IoU .25. The rank layer uses five fixed VisDrone configurations with identical 548/548 raw coverage: B-640, B-1280, Y11m, ASD, and Y11n-P2. At terminal contract S and each headline support, the pool-wide display reports global micro-F_1 at confidence .25, max-F_1 over the prespecified confidence grid .10–.50, and AP50 with a nontruncating terminal `maxDets`=2000. For the named Y11m–ASD pair, a 2×2 control crosses valid-GT IoU $\{.25, .50\}$ with max-F_1 and score-integrated AP; fixed-confidence F_1 is retained at each IoU as an operational endpoint. Neutralization remains fixed at intersection-over-prediction .5 in every cell, so changing localization does not also change ignore semantics.

VisDrone filenames identify 76 sequences. A paired cluster replicate samples 76 sequence identifiers with replacement, includes every image in each sampled sequence, applies identical multiplicities to both configurations, and recomputes pooled TP/FP/FN or pooled-detection AP. We use 10,000 replicates and percentile 95% intervals. Image-paired results are retained as a supplementary control. Finite-grid boundary cells are exploratory and are not used for confirmatory inference after inspection. An independent detection-record AP50 implementation is checked against COCOeval at the two candidates and two headline rules.

The ontology mappings, image denominators, support grid, contract order, headline rules, confidence grid, candidate pool, named top pair, AP cap, sequence definition, bootstrap seed, and replicate count are stored as machine-readable records. Only the two headline supports enter pairwise statements. The 2×2 control is restricted to that pair and those supports; it isolates metric factors rather than initiating a new configuration search. The larger 14-support-by-eight-confidence surface remains exploratory.

III. Results

A. Coverage and support qualification

Figure 1 shows full coverage for VisDrone (548/548) and UAVDT (305/305), but 1,869/2,804 images for AI-TOD. At 24 pixels, full-source retained support is 73.3%, 95.1%, and 9.7%; at .015 it is 89.8%, 97.4%, and 81.0%. Normalization narrows the three-source headline spread but does not remove it.

AI-TOD's missingness is support selective. Its covered images are 66.7% of the validation images but contain

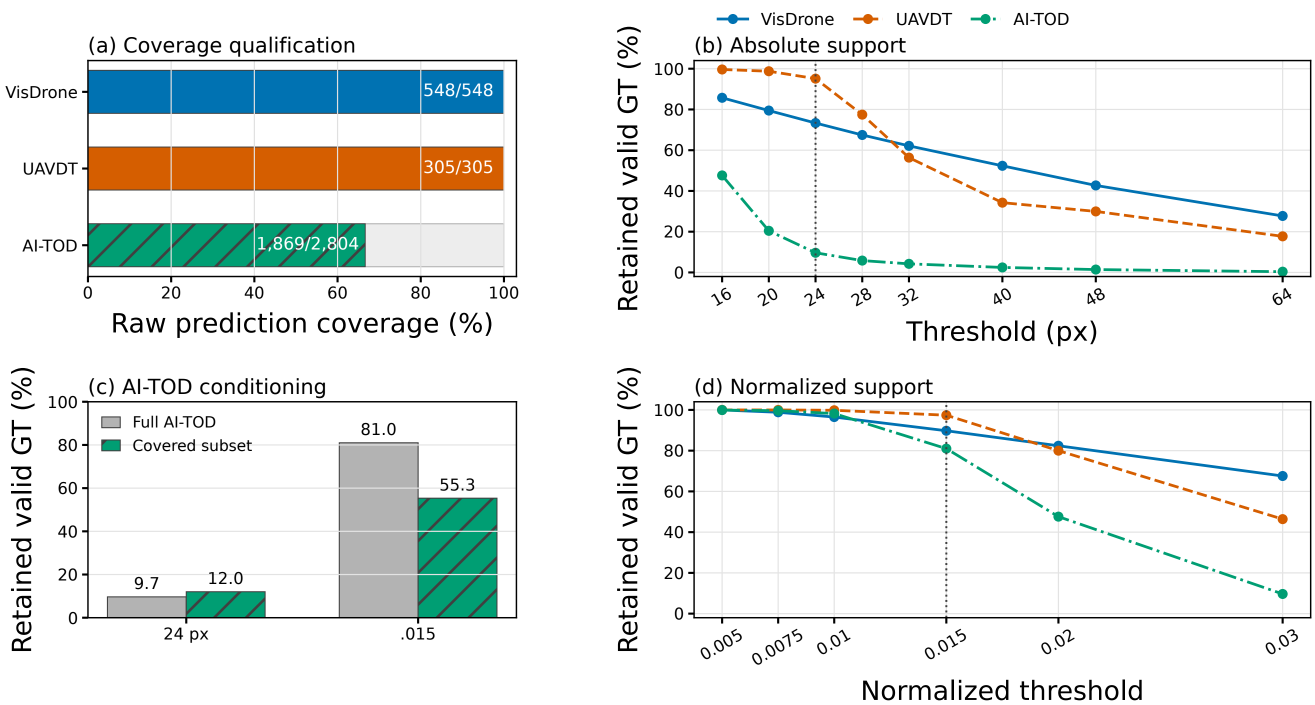

Fig. 1. Coverage qualification precedes metric interpretation. (a) Raw prediction coverage. (b) Full-source absolute retained-support curves. (c) AI-TOD's covered images contain 28.2% of its vehicle boxes; its full-source and covered-subset headline support therefore differ. (d) Full-source normalized retained-support curves. Percentages are pooled valid-GT fractions computed before prediction matching.

16,900/59,904=28.2% of vehicle boxes. Covered and noncovered images average 9.0 and 46.0 boxes; their medians are 0 and 21. The covered-subset retained fraction is 12.0% at 24 pixels and 55.3% at .015, versus 9.7% and 81.0% on the full annotations. AI-TOD therefore passes support diagnosis, but its metric values below describe only the declared covered subset.

The qualification is not triggered by coverage rate alone. Covered versus noncovered images have a boxes-per-image standardized mean difference of $-.673$; at box level, max-side Wasserstein distances are 2.96 pixels and .00370 in normalized units. Consequently, the uncovered part is neither treated as empty prediction nor repaired by weighting. The audit preserves two separate quantities: full-annotation support and covered-subset contract response.

B. Contract-path attribution

Figure 2 turns endpoint movement into an evaluator mechanism. At 24 pixels, the VisDrone path is $-.036, +.067, +.009$: filtering initially lowers F_1, removed-object neutralization more than recovers that loss, and source ignores add a smaller shift. The count ledger explains the middle step exactly: $F \rightarrow R$ changes no TP or FN but neutralizes 2,320 false positives. At .015, all three VisDrone steps are positive $(+.018, +.009, +.008)$, showing that the sign of target filtering itself depends on the declared support.

UAVDT moves little $(-.009, +.004, 0$ at 24 pixels and $-.002, +.001, 0$ at .015). AI-TOD behaves differently. Its absolute $A \rightarrow F$ change is $-.330$, followed by $+.080$ when 30,232 detections near removed objects become neutral; the terminal score remains .145 rather than returning to .395. At .015 the corresponding changes are $-.136$ and $+.066$. Thus the large AI-TOD endpoint range is not solely an artifact of counting detections on removed boxes as background: removed-object handling explains a substantial, separately measured component, while target contraction remains dominant.

Normalized support reduces, but does not eliminate, contract sensitivity in every source. The full four-contract range falls from .076 to .035 for VisDrone, from .0085 to .0023 for UAVDT, and from .330 to .136 for coverage-conditioned AI-TOD. The common direction is consistent with scale invariance, whereas the unequal residual ranges show why normalized support cannot substitute for a source-specific contract audit.

C. Metric- and sequence-qualified ranking

At IoU .25, the fixed operating point favors Y11m over ASD at both rules: the absolute difference is .0229 [.0185,.0275], and the normalized difference is .0158 [.0118,.0201]. Reoptimizing confidence preserves the direction, with max-F_1 differences .0163 [.0119,.0213] and .0099 [.0060,.0140]. At IoU .50, fixed-F_1 remains positive at .0222 [.0180,.0269] and .0141 [.0104,.0181], as does max-F_1 at .0141 [.0099,.0191] and .0054 [.0013,.0103]. Neither confidence choice nor stricter localization alone reverses the pair.

Figure 3(b) separates score integration from localization. At IoU .25, AP differences are +.0081 [.0051,.0120] at 24 pixels and $-.0005$ [$-.0030$,.0047] at .015. AP50 then gives +.0013 [$-.0033$,.0052] and $-.0110$ [$-.0181$,$-.0057$]. Thus

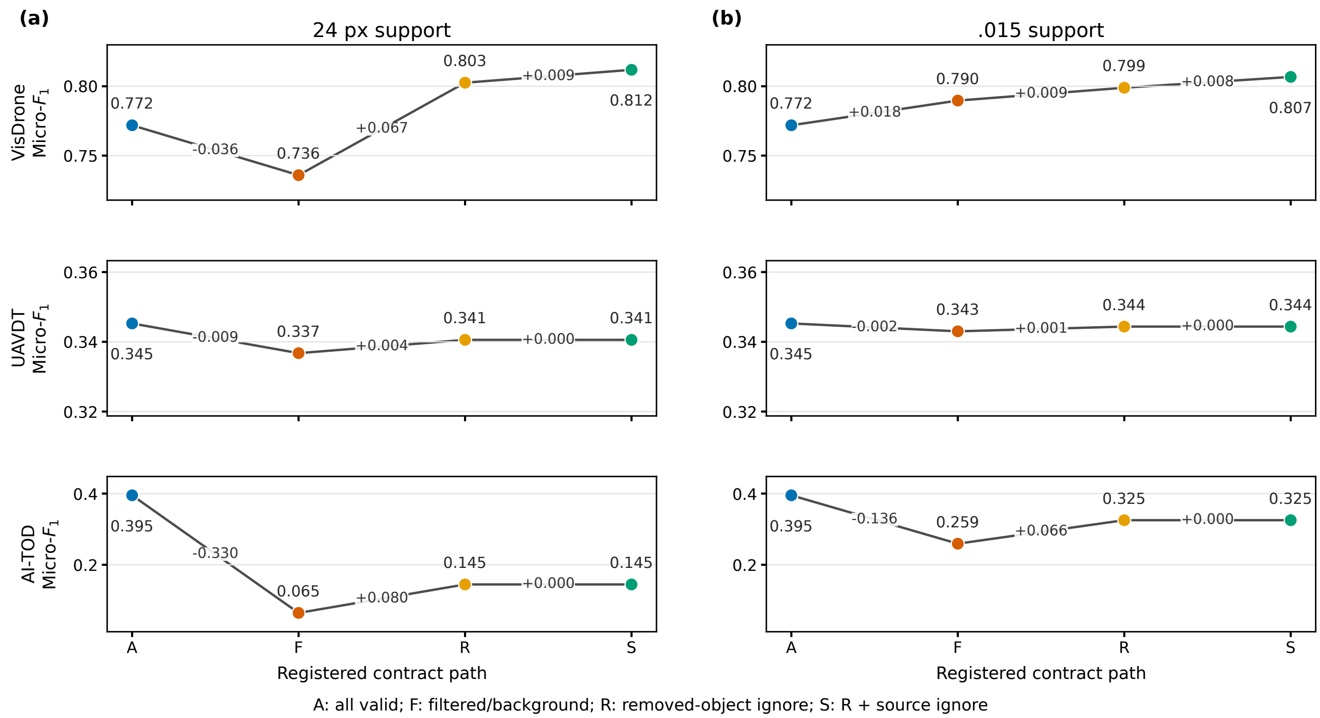

Fig. 2. Registered $A \to F \to R \to S$ micro-F_1 paths for one fixed baseline artifact per source at confidence and IoU .25. (a) Absolute 24-pixel support. (b) Normalized .015 support. Adjacent labels are exact path changes; UAVDT and AI-TOD have no active source-ignore regions, hence $R = S$. AI-TOD values are conditioned on 1,869 covered images.

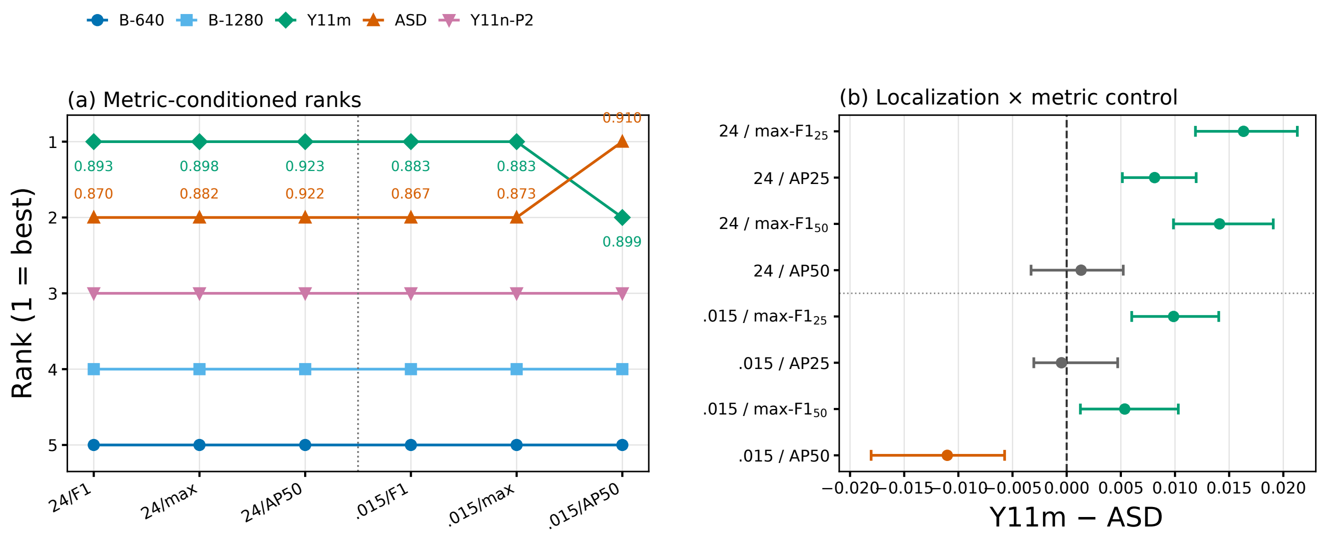

Fig. 3. Metric-qualified ranking under terminal contract S on the common 548-image VisDrone set. (a) Pool-wide ranks and Y11m/ASD point values for fixed-F_1, max-F_1, and AP50. (b) Y11m–ASD sequence-paired 95% intervals for the 2×2 localization-by-metric control; green, orange, and gray denote intervals above, below, and crossing zero. AP is recomputed from pooled detections in each of 10,000 resamples of 76 sequences.

score integration at loose localization preserves the 24-pixel advantage and makes the normalized comparison unresolved; strict localization alone still favors Y11m. A sequence-resolved ASD-favoring normalized result appears only in the combined AP50 cell. At 24 pixels the same combination collapses, but does not reverse, the margin. The custom AP50 values match COCOeval at all four endpoints.

D. Evaluator controls

The declared AP cap is 2,000 detections per image, while the largest observed candidate output is 1,881; headline AP therefore contains no cap-truncated image. This matters because caps of 100 or 300 alter dense aerial-image comparisons even when the winner does not change. For the two candidates and two headline contracts, the independent score-ordered

AP calculation and COCOeval agree to machine precision. A separate threshold-first Hungarian replay on the released matching-control subset agrees with greedy matching on the selected winner in all 24 multi-candidate settings; its largest absolute F_1 difference is .00546. The matching control validates an aligned component rather than extending the headline rank claim.

Image-paired controls preserve the same endpoint directions. Sequence intervals remain primary because adjacent VisDrone frames are not exchangeable independent observations; all image-level intervals are tabulated in the released records.

IV. Discussion

The gate, path, ledger, and cluster interval answer different questions. Coverage determines which population is observed. Retained support states which valid target remains. The ordered path attributes deterministic metric movement to one evaluator change at a time. Factorial metric controls and paired intervals then identify which metric factors change a named comparison on common sequences. A support shift is not sampling uncertainty, and a resolved F_1 margin is not automatically an AP conclusion.

We therefore qualify a rank statement along three independent axes. It is *coverage qualified* only on an explicit common image set, *metric qualified* only for the declared localization and score treatment, and *sampling qualified* only relative to the prespecified resampling unit. Here Y11m's VisDrone advantage persists for fixed- and optimized-confidence F_1 at both IoUs. Score integration weakens that conclusion, and only its combination with IoU .50 produces a sequence-resolved ASD-favoring normalized endpoint. AI-TOD endpoints are contract qualified on covered images but not coverage qualified for the full benchmark.

The path is an audit coordinate system, not a claim that S is universally correct. Applications may choose whether removed valid objects should be background or neutral, but that choice should be explicit and its consequence reported. Likewise, resize invariance makes normalized support comparable under uniform scaling; it does not remove source semantics or acquisition differences. The three-source results empirically demonstrate both points.

The ordered path supplies information that an endpoint band cannot. Comparing A with S alone would merge a target change, a removed-object decision, and native ignore semantics into one number. The intermediate counts show which mechanism dominates and whether a recovery is caused by fewer FN, by neutralized FP, or by both. This attribution is especially important when a support cutoff removes many labeled objects, because a large score response can otherwise be mistaken for model instability or annotation error.

The practical output is a compact audit record rather than a preferred universal threshold. It contains the annotation denominator and prediction coverage; retained support in absolute and normalized coordinates; contract endpoints and integer count transitions; the exact common set and metric definition used for each comparison; and paired uncertainty at the source's dependence unit. These fields make two studies comparable even when they choose different valid application contracts. They also expose when an apparent robustness claim comes from a saturated detection cap, an incomplete prediction cache, a fixed confidence calibration, or image-wise resampling of correlated frames.

The evidence remains artifact scoped. AI-TOD has no full-benchmark metric or rank inference; VisDrone conclusions concern five frozen configurations; sequence clusters are derived from filenames; and no transition estimates annotation-error rates or causal detector effects. Full threshold records, the 18-row count ledger, the metric-factorial control, coverage diagnostics, image-paired intervals, matching and cap checks, and replay instructions appear in the supplementary material and released repository: https://github.com/Marchematics/aerial-evaluation-audit.

V. Conclusion

Coverage-qualified contract auditing converts a support-rule sensitivity check into a reproducible evaluation method. It identifies the observed population, measures support, decomposes target and ignore semantics with exact count transitions, and separates localization, score integration, and sequence uncertainty in common-coverage ranks. The resulting record explains large score movement without overstating a detector winner.

References

[1] A. Torralba and A. A. Efros, "Unbiased look at dataset bias," in *CVPR 2011*. IEEE, 2011, pp. 1521–1528.

[2] X. Zhao, S. Schulter, G. Sharma, Y.-H. Tsai, M. Chandraker, and Y. Wu, "Object detection with a unified label space from multiple datasets," in *European conference on computer vision*. Springer, 2020, pp. 178–193.

[3] C. G. Northcutt, A. Athalye, and J. Mueller, "Pervasive label errors in test sets destabilize machine learning benchmarks," *arXiv preprint arXiv:2103.14749*, 2021.

[4] D. Bolya, S. Foley, J. Hays, and J. Hoffman, "Tide: A general toolbox for identifying object detection errors," in *European Conference on Computer Vision*. Springer, 2020, pp. 558–573.

[5] F. C. Akyon, S. O. Altinuc, and A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," in *2022 IEEE international conference on image processing (ICIP)*. IEEE, 2022, pp. 966–970.

[6] E. Zhu, Z. Chen, D. Wang, H. Shi, X. Liu, and L. Wang, "UNetMamba: An efficient UNet-like Mamba for semantic segmentation of high-resolution remote sensing images," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, pp. 1–5, 2025.

[7] G. Liu, J. Wen, D. You, Y. Tang, Y. Han, and Q. Li, "Impacts of DEM geolocation bias on multiscale validation of land surface albedo over rugged terrain," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, pp. 1–5, 2025.

[8] P. Zhu, L. Wen, X. Bian, H. Ling, and Q. Hu, "Vision meets drones: A challenge," *arXiv preprint arXiv:1804.07437*, 2018.

[9] D. Du, Y. Qi, H. Yu, Y. Yang, K. Duan, G. Li, W. Zhang, Q. Huang, and Q. Tian, "The unmanned aerial vehicle benchmark: Object detection and tracking," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 370–386.

[10] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Detecting tiny objects in aerial images: A normalized wasserstein distance and a new benchmark," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 190, pp. 79–93, 2022.

[11] D. Lam, R. Kuzma, K. McGee, S. Dooley, M. Laielli, M. Klaric, Y. Bulatov, and B. McCord, "xview: Objects in context in overhead imagery," *arXiv preprint arXiv:1802.07856*, 2018.

[12] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *European conference on computer vision*. Springer, 2014, pp. 740–755.